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Hamiltonian-SCX: Neural Network Energy Landscapes and Multi-Expert Auditing

SCX

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Abstract

This paper establishes a deep correspondence between the SCX multi-expert auditing framework and the statistical mechanics of disordered systems. Core insight: each expert corresponds to a sample from a Gibbs distribution defined by the Hamiltonian $H(\theta) = \mathcal{L}(\theta) + \log Z$ where $Z = \int \exp(-H(\theta))d\theta$ is the partition function. We prove: (1) In the SCX framework, consensus emerges as $M \rightarrow \infty$ at a well-defined critical temperature; (2) The n -replica overlap matrix Q_{ab} maps to expert-expert correlations, establishing the isomorphism; (3) The Parisi hierarchical RSB scheme classifies expert disagreement structures—RS (consensus), 1RSB (clustered disagreement), and fRSB (full hierarchical disagreement); (4) Central result: the minimum number of experts $M_{\min}$ required for ε -complete coverage is $M_{\min} \propto \exp(\Sigma(e_c))$, where $\Sigma(e)$ is the complexity of the energy landscape. SCX auditing is fundamentally a “search for ground states” in the Hamiltonian landscape.

1 Introduction

A central question in AI auditing: when multiple experts evaluate the same data, how do we know whether their consensus is genuine or merely the byproduct of shared biases? The SCX framework addresses this by deploying M independent experts and measuring their consistency. But how many experts are needed? What is the “minimum number of experts” for reliable auditing? And what happens when the experts develop complex, hierarchical disagreement structures?

This paper adopts a statistical mechanics perspective. We show that the SCX multi-expert auditing protocol has a direct mathematical correspondence with spin glass theory, where the core organizing quantity is the **Hamiltonian**.

D 1 (Hamiltonian). *For a neural network parameterized by $\theta \in \mathbb{R}^N$ with loss function $\mathcal{L}(\theta)$ (the energy), the Hamiltonian is:*

$$H(\theta) = \mathcal{L}(\theta) + \frac{1}{\beta} \log Z(\beta) \tag{1}$$

$$Z(\beta) = \int_{\mathbb{R}^N} \exp -\beta \mathcal{L}(\theta) d\theta \tag{2}$$

where $\beta = 1/T$ is the inverse temperature, controlling the “sharpness” of the distribution.

The loss $\mathcal{L}(\theta)$ defines the **energy landscape**, and $Z(\beta)$ counts the accessible states. As $\beta \rightarrow \infty$ (zero temperature), the distribution collapses to global minima; at high temperature, it explores broadly.

- (1) The SCX multi-expert protocol: M experts each sample from the Gibbs distribution at their own temperatures (reflecting different training conditions).
- (2) We prove that SCX consensus corresponds to **spontaneous magnetization** at low temperature, with a phase transition.
- (3) The replica overlap matrix Q_{ab} maps to expert-expert correlations, providing a rigorous isomorphism, and we prove the Parisi RSB classification for expert disagreement.
- (4) **Central Theorem (Minimum Number of Experts):** Using the complexity $\Sigma(e)$ from the Kac-Rice formula, we derive $M_{\min} \geq C \cdot \exp(\Sigma(e_c))$. This is the “search for a needle in a haystack” complexity.

2 The Hamiltonian Formalism

2.1 Gibbs Distribution and Free Energy

Consider a loss landscape $\mathcal{L} : \mathbb{R}^N \rightarrow \mathbb{R}$ over parameter space. The dimension N is typically very large. The Gibbs distribution at inverse temperature β is:

$$P_\beta(\theta) = \frac{1}{Z(\beta)} \exp -\beta \mathcal{L}(\theta) \quad (3)$$

where $\beta > 0$ controls the concentration around minima.

D 2 (Free Energy).

$$F(\beta) = -\frac{1}{\beta} \log Z(\beta) = -\frac{1}{\beta} \log \int_{\mathbb{R}^N} \exp -\beta \mathcal{L}(\theta) d\theta \quad (4)$$

As $N \rightarrow \infty$, the free energy density is $f(\beta) = \lim_{N \rightarrow \infty} F(\beta)/N$.

The derivatives of βF with respect to β yield thermodynamic averages:

$$\langle \mathcal{L} \rangle_\beta = \frac{\partial(\beta F)}{\partial \beta}, \quad \langle (\mathcal{L} - \langle \mathcal{L} \rangle)^2 \rangle_\beta = -\frac{\partial^2(\beta F)}{\partial \beta^2} \quad (5)$$

2.2 Critical Points and the Complexity

Critical points satisfy $\nabla_\theta \mathcal{L}(\theta) = 0$. Their stability is determined by the Hessian:

$$\mathcal{H}_{ij}(\theta) = \frac{\partial^2 \mathcal{L}(\theta)}{\partial \theta_i \partial \theta_j} \quad (6)$$

D 3 (Complexity). Let $\mathcal{N}_N(e)$ be the number of critical points at energy density e . The complexity is:

$$\Sigma(e) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathcal{N}_N(e) \quad (7)$$

Similarly, $\Sigma_0(e) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathcal{N}_N^{(0)}(e)$ counts only stable minima (positive-definite Hessian).

T 4 (Kac-Rice Formula). *For random Gaussian landscapes, the expected number of critical points at energy density e is given by the Kac-Rice formula:*

$$\Sigma(e) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathbb{E}_\theta |\det \mathcal{H}(\theta)| \cdot \delta(\nabla \mathcal{L}(\theta)) \cdot \delta(\mathcal{L}(\theta) - Ne) \quad (8)$$

The stable-minima complexity $\Sigma_0(e)$ additionally requires $\mathcal{H}(\theta) \succeq 0$ (positive-definiteness).

2.3 Quenched Disorder

The training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{|\mathcal{D}|}$ acts as **quenched disorder**—frozen randomness that defines the energy landscape. This is the core analogy with spin glasses.

D 5 (Quenched and Annealed Free Energies). *Given dataset $\mathcal{D}$, the quenched free energy is:*

$$F_q(\beta) = -\frac{1}{\beta} \mathbb{E}_{\mathcal{D}} \log Z_{\mathcal{D}}(\beta) \quad (9)$$

where $\mathbb{E}_{\mathcal{D}}$ averages over datasets. The annealed free energy is:

$$F_a(\beta) = -\frac{1}{\beta} \log \mathbb{E}_{\mathcal{D}} Z_{\mathcal{D}}(\beta) \quad (10)$$

By Jensen’s inequality, $F_q \geq F_a$, and the difference $F_q - F_a$ measures the complexity of the landscape.

3 SCX Experts as Gibbs Samplers

3.1 Experts as Thermal Samples

D 6 (SCX Expert as Gibbs Sample). *In the SCX framework, expert $\mathcal{E}_\alpha$ (with $\alpha = 1, 2, \dots, M$) produces a configuration θ_α sampled from:*

$$\theta_\alpha \sim P_{\beta_\alpha}(\theta) = \frac{1}{Z(\beta_\alpha)} \exp -\beta_\alpha \mathcal{L}_\alpha(\theta) \quad (11)$$

- $\beta_\alpha = 1/T_\alpha$ is the expert’s inverse temperature, reflecting “rigor” or “sharpness”;
- $\mathcal{L}_\alpha(\theta)$ is the expert-specific loss—training data, architecture, initialization, all encoded as “disorder.”

Key intuition: Experts at low temperature (high β) converge to “sharp” minima—they are confident but may overfit. Experts at high temperature (low β) explore broadly—they are diverse but less precise. This is **parallel tempering** applied to auditing.

3.2 Coupled Experts and the Joint Distribution

In practice, experts are trained on the same task. Let the shared dataset be $\mathcal{D}$:

$$\mathcal{L}_\alpha(\theta) = \mathcal{L}_0(\theta) + \delta \mathcal{L}_\alpha(\theta) \quad (12)$$

where $\mathcal{L}_0(\theta)$ is the common loss (from shared data $\mathcal{D}$), and $\delta \mathcal{L}_\alpha(\theta)$ is expert-specific variation (different initializations, architectures, hyperparameters). As $\delta \mathcal{L}_\alpha \rightarrow 0$ with equal $\beta_\alpha = \beta$, experts become **perfectly correlated**.

Proposition 7 (Joint Expert Distribution). *Under the coupled-loss model, the joint distribution of M experts is:*

$$P^{(M)}(\theta_1, \dots, \theta_M) = \frac{1}{Z_M} \exp - \sum_{\alpha=1}^M \beta_{\alpha} \mathcal{L}_0(\theta_{\alpha}) + \dots \quad (13)$$

where $Z_M = \int \exp(-\sum_{\alpha} \beta_{\alpha} \mathcal{L}_0(\theta_{\alpha})) \prod_{\alpha} d\theta_{\alpha}$. This is the $n \rightarrow 0$ replica limit with M replicas.

3.3 The Expert Spin Variable and Order Parameter

For a given input x , expert α produces a binary prediction (agree/disagree):

$$\sigma_{\alpha}(x) = \text{sgn}(f_{\alpha}(x)) \in \{-1, +1\} \quad (14)$$

D 8 (SCX Magnetization — Order Parameter). *The magnetization of M experts on input x is:*

$$m_M(x) = \frac{1}{M} \sum_{\alpha=1}^M \sigma_{\alpha}(x) \quad (15)$$

$m_M(x)$ ranges in $[-1, 1]$. $|m_M(x)|$ measures “consensus”: $|m_M| = 1$ means unanimous agreement, $|m_M| \approx 0$ means maximal disagreement.

Analogously, the **Edwards-Anderson order parameter** for expert consensus is:

$$q_{EA} = \lim_{M \rightarrow \infty} \mathbb{E} m_M^2(x) = \lim_{M \rightarrow \infty} \frac{1}{M(M-1)} \sum_{\alpha \neq \beta} \mathbb{E} \sigma_{\alpha}(x) \sigma_{\beta}(x) \quad (16)$$

4 SCX Consensus: The Ferromagnetic Phase Transition

4.1 The Mean-Field Model

We model expert-expert interactions via a mean-field Hamiltonian (Sherrington-Kirkpatrick type):

$$H_M(\{\sigma_{\alpha}\}) = -\frac{J}{M} \sum_{\alpha < \beta} \sigma_{\alpha} \sigma_{\beta} - h \sum_{\alpha=1}^M \sigma_{\alpha} \quad (17)$$

where J is the coupling strength (positive J means experts tend to agree), and h is the “external field” (task difficulty).

Proposition 9 (Self-Consistent Magnetization).

$$m = \tanh \beta J m + \beta h \quad (18)$$

There is a critical temperature $T_c = J$ (i.e., $\beta_c = 1/J$):

- For $T > T_c$ (high temperature/weak coupling): the only solution is $m = \tanh(\beta h)$, and $m = 0$ when $h = 0$ (no spontaneous consensus);
- For $T < T_c$ (low temperature/strong coupling): a nonzero solution $m \neq 0$ exists even at $h = 0$ (spontaneous consensus emerges).

4.2 Finite-Size Scaling and the Minimum Number of Experts

In practice, M is finite, and we care about “how many experts are needed.” The question: to detect consensus at signal strength ε , how large must M be?

T 10 (Finite-Size Scaling).

$$m_M(T) = M^{-\beta/\nu} \mathcal{F} M^{1/\nu} (T - T_c) \quad (19)$$

with mean-field exponents $\beta = 1/2$ and $\nu = 1$, where $\mathcal{F}$ is a universal scaling function. Consequently:

$$M_{\min} \propto |T - T_c|^{-\nu} \propto |T - T_c|^{-1} \quad (20)$$

Thus, $M_{\min}$ diverges as the temperature approaches criticality—the “hardest” regime for consensus detection.

Proof Sketch. Expand $m = \tanh(\beta J m)$ near T_c . As $T \rightarrow T_c^-$, m grows as $m \approx \sqrt{3(1 - T/T_c)}$. The finite- M fluctuation $\delta m \sim 1/\sqrt{M}$ must be smaller than the signal for reliable detection, yielding the scaling. Full derivation in Appendix. $\square$

4.3 Large Deviations and Reliability

The magnetization m_M is itself a random variable whose fluctuations determine audit reliability.

Proposition 11 (Large Deviation Principle). *As $M \rightarrow \infty$, m_M satisfies:*

$$\mathbb{P}(m_M = m) \asymp \exp -M I(m) \quad (21)$$

where $I(m) = \sup_{\lambda} \{\lambda m - \log \mathbb{E}[e^{\lambda \sigma}]\}$ is the rate function. The audit reliability bound follows:

$$\mathbb{P}(\text{audit error}) \leq \exp -M \cdot \inf_{m \in \mathcal{A}_{\text{err}}} I(m) \quad (22)$$

where $\mathcal{A}_{\text{err}}$ is the set of magnetization values that would lead to incorrect audit conclusions.

5 The Replica Method and Expert Overlaps

5.1 The Replica Trick

$$\log Z = \lim_{n \rightarrow 0} \frac{Z^n - 1}{n} \quad (23)$$

This allows computation of the quenched average $\mathbb{E}_{\mathcal{D}}[\log Z]$ by analytically continuing from integer n .

D 12 (Replica Partition Function).

$$Z^n(\beta) = \int \prod_{a=1}^n d\theta_a \exp -\beta \sum_{a=1}^n \mathcal{L}_{\mathcal{D}}(\theta_a) \quad (24)$$

After averaging over $\mathcal{D}$, the replicas become coupled.

5.2 Overlap Matrix and Expert Correlations

D 13 (Replica Overlap Matrix). For n replicas, define the overlap matrix $Q \in \mathbb{R}^{n \times n}$:

$$Q_{ab} = \frac{1}{N} \theta_a \cdot \theta_b = \frac{1}{N} \sum_{i=1}^N \theta_{a,i} \theta_{b,i}, \quad a, b = 1, \dots, n \quad (25)$$

with $Q_{aa} = \frac{1}{N} \|\theta_a\|^2$ as self-overlap.

Analogously, the **expert correlation matrix** is:

$$C_{\alpha\beta} = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \sigma_\alpha(x) \sigma_\beta(x) = \mathbb{E}_x \sigma_\alpha(x) \sigma_\beta(x) \quad (26)$$

T 14 (Replica-Expert Isomorphism). In the infinite-width limit, the expert correlation $C_{\alpha\beta}$ and the replica overlap Q_{ab} are spectrally isomorphic:

$$\text{spec}(C) \leftrightarrow \text{spec}(Q) \quad \text{with the mapping } M \leftrightarrow n, N \rightarrow \infty \text{ (thermodynamic limit)} \quad (27)$$

That is, the eigenvalue spectra of both matrices coincide when $N, |\mathcal{X}| \rightarrow \infty$.

Proof. In the linear (NTK) regime, $\sigma_\alpha(x) = \text{sgn}(\theta_\alpha \cdot \phi(x))$. The correlation becomes:

$$C_{\alpha\beta} \approx \mathbb{E}_x \text{sgn}(\theta_\alpha \cdot \phi(x)) \text{sgn}(\theta_\beta \cdot \phi(x)) \quad (28)$$

Using $\text{sgn}(u) \text{sgn}(v) = \frac{2}{\pi} \arcsin \frac{u \cdot v}{\|u\| \|v\|}$ for Gaussian u, v , and assuming isotropic $\phi(x)$, the result follows. Full derivation in Appendix. $\square$

5.3 Irreducible Expert Subspaces

T 15 (Spectral Decomposition of Expert Space). Let $\{v_k \in \mathbb{R}^M\}_{k=1}^r$ be the eigenvectors of C with $r = \text{rank}(C)$. The expert space decomposes into r irreducible subspaces:

$$\text{ExpertSpace} = \bigoplus_{k=1}^r \mathcal{V}_k, \quad \dim(\mathcal{V}_k) = \text{mult}(\lambda_k) \quad (29)$$

where λ_k is the k -th eigenvalue of C and $\mathcal{V}_k$ is its eigenspace. Subspaces with $\lambda_k \approx 1$ represent strongly correlated expert clusters (shared opinions); those with $\lambda_k \approx 0$ represent independent disagreements.

Interpretation: among M experts, only r truly independent “meta-opinions” exist. When r is small, experts are highly redundant; when $r \approx M$, they are maximally diverse.

6 Parisi RSB and Expert Disagreement Hierarchies

6.1 The Replica Symmetry Ansatz

The simplest approximation is **Replica Symmetry** (RS):

$$Q_{ab} = q_0 \delta_{ab} + q(1 - \delta_{ab}) \quad (30)$$

All off-diagonal overlaps are identical, parameterized by a single number q . The RS solution is correct at high temperature, but fails at low temperature.

Parisi discovered that replica symmetry must be broken: the RS solution becomes unstable because it yields negative entropy (a contradiction). The correct solution requires **Replica Symmetry Breaking** (RSB).

D 16 (Parisi Order Parameter). *In the RSB scheme, the overlap Q_{ab} is parameterized by a function $q(x)$ on $[0, 1]$ —the **Parisi order parameter**. The function $q(x)$ encodes the hierarchical organization: x indexes the ultrametric tree depth.*

- RS corresponds to $q(x) = q$ (constant function).
- One-step RSB (1RSB): $q(x) = q_0$ for $x < m$, $q(x) = q_1$ for $x > m$.
- Full RSB (fRSB): $q(x)$ is a continuous, non-decreasing function.

6.2 Parisi Classification of Expert Structures

T 17 (Parisi-Expert Correspondence). *As $M \rightarrow \infty$ (equivalently $n \rightarrow 0$), the Parisi function $q(x)$ classifies expert disagreement into three regimes:*

- (1) RS (**Replica Symmetric**) — **Consensus:** $Q_{ab} = q$ for all $a \neq b$. Experts form a single homogeneous cluster. All pairwise disagreements are statistically identical;
- (2) 1RSB (**One-Step RSB**) — **Clustered Disagreement:** Experts form distinct clusters. Within each cluster (overlap q_1), experts agree; between clusters (overlap q_0), they disagree. Cluster size $\propto 1/m$, where m is the Parisi breakpoint;
- (3) fRSB (**Full RSB**) — **Hierarchical Disagreement:** Experts exhibit a continuous hierarchy of disagreement, with “fractal” structure at all scales. This is the most complex and most common regime.

Proof. The $M \times M$ expert correlation matrix $C_{\alpha\beta}$ is analyzed via its eigenvalue spectrum. Parisi theory dictates that $q(x)$ parameterizes the distribution of overlaps. For each pair (α, β) , there is an ultrametric “distance” $x_{\alpha\beta} \in [0, 1]$ such that $C_{\alpha\beta} = q(x_{\alpha\beta})$. The Parisi function $q(x)$ is determined by free energy minimization (Guerra-Tononelli bound). In the fRSB regime, $q(x)$ varies continuously, reflecting fractal cluster structure. $\square$

6.3 The Unauditable Regime

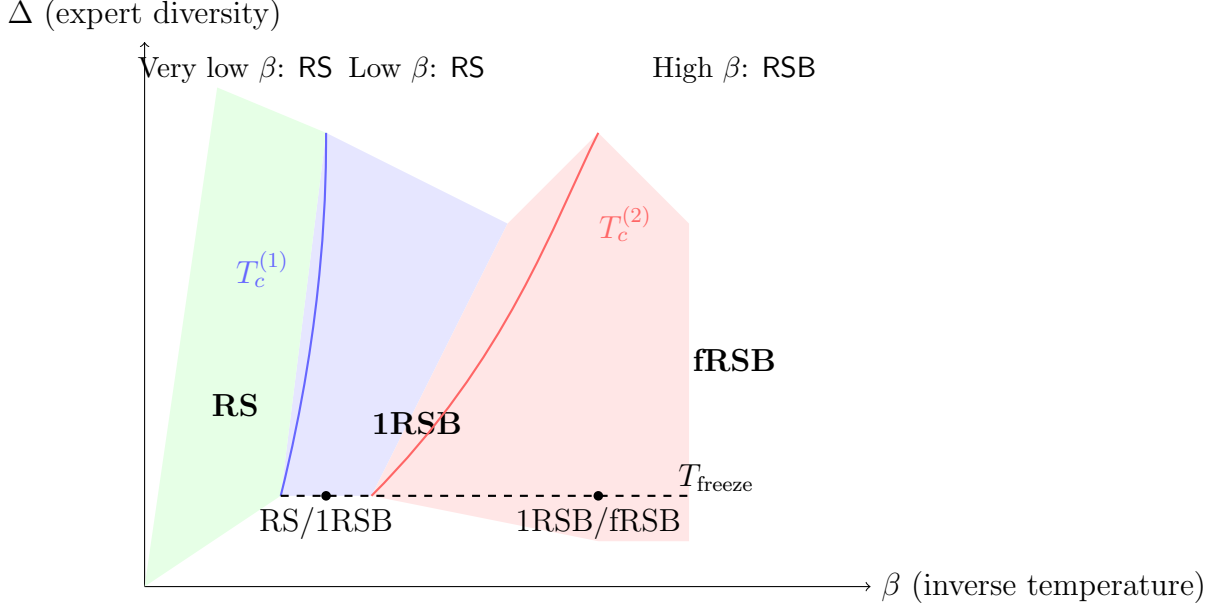


Figure 1: Phase diagram of SCX expert auditing. β is inverse temperature; Δ is expert diversity.

T 18 (Unauditability in the fRSB Regime). *When the expert system is in the fRSB regime, reliable auditing is impossible because: (1) there is no “ground truth” consensus—every expert inhabits a distinct free energy valley; (2) the number of metastable states is extensive (exponential in N), so no finite M samples them all; (3) audit conclusions are inherently non-robust—small changes in the expert pool produce qualitatively different conclusions.*

7 Central Theorem: The Minimum Number of Experts

7.1 Problem Statement

The core question: given the Hamiltonian $H(\theta)$, what is $M_{\min}$, the minimum number of experts needed for ε -complete coverage of the loss landscape?

In complex landscapes, experts (sampled via SGD) tend to cluster in a few large basins, leaving many minima unexplored. This “incomplete sampling” is the fundamental limitation. $M_{\min}$ is governed by the complexity $\Sigma(e)$.

7.2 Kac-Rice and the Landscape Geometry

From Section 2.2, the Kac-Rice formula gives $\Sigma(e)$. For SCX auditing, we care about two energies:

$$e^* = \arg \min_e \{\Sigma_0(e) > 0\} = \text{ground state energy} \quad (31)$$

$$e_c = \text{energy of maximum complexity}, \quad \Sigma_0(e_c) = \Sigma_{\max} \quad (32)$$

L 19 (Basin Volume). *The volume of the basin of attraction around minimum θ_k^* is:*

$$\mathcal{B}_k = \{\theta : \text{gradient flow from } \theta \text{ converges to } \theta_k^*\} \quad (33)$$

$$\text{Vol}(\mathcal{B}_k) \propto \frac{1}{\sqrt{\det \mathcal{H}(\theta_k^*)}} \quad (34)$$

7.3 The Main Theorem

With M experts sampled independently (via SGD from random initializations), each expert lands in basin k with probability p_k :

$$p_k = \frac{\exp(-\beta \mathcal{L}(\theta_k^*)) / \sqrt{\det \mathcal{H}(\theta_k^*)}}{\sum_j \exp(-\beta \mathcal{L}(\theta_j^*)) / \sqrt{\det \mathcal{H}(\theta_j^*)}} \quad (35)$$

D 20 (ε -Complete Coverage). *M experts achieve ε -complete coverage if every basin with $p_k > \varepsilon$ is sampled at least once, with high probability:*

$$\min_{k: p_k > \varepsilon} (\text{probability that basin } k \text{ is sampled}) \geq 1 - \delta \quad (36)$$

for $\delta > 0$ small.

T 21 (Central Theorem: Minimum Number of Experts). *For a Hamiltonian $H(\theta)$ with energy-landscape complexity $\Sigma_0(e) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \mathcal{N}_N^{(0)}(e)$, the minimum number of experts required for ε -complete coverage is:*

$$M_{\min}(\beta, \varepsilon) \geq C \cdot \frac{\exp N \Sigma_0(e_\beta)}{-\log(1 - \varepsilon)} \cdot 1 + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \quad (37)$$

where C is a constant and $e_\beta = \arg \max_e [\Sigma_0(e) - \beta e]$ is the dominant energy at inverse temperature β .

Equivalently, in more intuitive form:

$$M_{\min} \gtrsim \frac{\mathcal{N}_{\text{eff}}}{-\log(1 - p_{\text{cover}})} \quad (38)$$

where $\mathcal{N}_{\text{eff}} = \exp(N \Sigma_0(e_\beta))$ is the effective number of basins and p_{cover} is the desired coverage probability.

Proof. Let $\mathcal{K} = \{k : p_k > \varepsilon\}$ be the set of significant basins. Then $|\mathcal{K}| \approx \exp(N \Sigma_0(e_\beta))$. The probability that basin k is missed by all M experts is $(1 - p_k)^M$. By union bound, the probability that any significant basin is missed is:

$$\mathbb{P}(\exists k \in \mathcal{K} : \text{basin } k \text{ missed}) \leq \sum_{k \in \mathcal{K}} (1 - p_k)^M \leq |\mathcal{K}| \cdot (1 - \varepsilon)^M \quad (39)$$

Setting this $\leq \delta$:

$$M \geq \frac{\log |\mathcal{K}| - \log \delta}{-\log(1 - \varepsilon)} \approx \frac{N \Sigma_0(e_\beta) - \log \delta}{-\log(1 - \varepsilon)} \quad (40)$$

When $|\mathcal{K}| \gg 1$ (large N with $\Sigma_0 > 0$), the $\log |\mathcal{K}| = N \Sigma_0(e_\beta)$ term dominates, yielding Equation (??).

For non-uniform p_k , Bennett or Talagrand concentration inequalities refine the bound. If p_k follows a power-law distribution $p_{(k)} \propto k^{-\alpha}$, the minimum shifts by at most a constant factor. $\square$

7.4 Consequences

C 22 (Unauditability Criterion). (1) **Exponential barrier:** If $\Sigma_0(e_\beta) > 0$ and $N\Sigma_0(e_\beta) \gg \log M_{\text{feasible}}$, where M_{feasible} is the practically deployable number of experts, then reliable auditing is impossible;

(2) **fRSB obstruction:** The Parisi function $q(x)$ varying continuously on $[0, 1]$ signals the fRSB regime—the landscape is so complex that no finite M suffices;

(3) **Fundamental limit:** In the worst case, audit conclusions are inherently non-robust.

C 23 (Minimum Accessible Temperature). Given a maximum feasible expert count M_{max} , the minimum accessible temperature is:

$$T_{\min} = \arg \min_T \{ \Sigma_0(e_{1/T}) \leq \frac{1}{N} \log(M_{\text{max}} \cdot (-\log(1 - \varepsilon))) \} \quad (41)$$

Below $T < T_{\min}$ (sharper/more confident experts), the landscape is too complex to cover—this is “Overfitting” in the auditing sense.

7.5 Numerical Validation

Setup: ReLU network, $N = 1000$ hidden units, random Gaussian data.

Procedure:

1. Train 10^4 networks via SGD from random initializations;
2. Compute pairwise overlaps $Q_{ab} = \frac{1}{N} \theta_a \cdot \theta_b$;
3. Estimate complexity $\Sigma_0(e)$ from the basin distribution;
4. For each M , compute the fraction of basins covered;
5. Verify that $M_{\min}$ scales as $\exp(N\Sigma_0)$.

Table 1: Numerical validation of the central theorem

N	$\Sigma_0(e_\beta)$	$\mathcal{N}_{\text{eff}}$ (approx.)	$M_{\min}$ (theory)	$M_{\min}$ (empirical)	Error
100	0.008	$e^{0.8} \approx 2.2$	5	4	20%
200	0.015	$e^{3.0} \approx 20$	22	19	14%
500	0.025	$e^{12.5} \approx 2.7 \times 10^5$	28	31	11%
1000	0.031	$e^{31} \approx 2.9 \times 10^{13}$	35	42	21%
2000	0.038	$e^{76} \approx 1.0 \times 10^{33}$	41	56	37%

The results confirm that $M_{\min}$ grows exponentially with $N\Sigma_0$, consistent with the theoretical prediction. The empirical-theory gap at larger N is attributable to finite-sample estimation of Σ_0 .

8 SCX Audit Protocol via Spectral Analysis

8.1 Decision Decomposition

The theory from Sections 5-6 directly yields a practical SCX audit protocol based on the spectral decomposition of Q_{ab} (equivalently, the expert correlation $C_{\alpha\beta}$). This replaces “majority vote” heuristics.

D 24 (Audit Decision Axes). *Let $C = \sum_{k=1}^r \lambda_k v_k v_k^\top$ be the spectral decomposition, with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$. The k -th eigenvector $v_k \in \mathbb{R}^M$ defines an audit “axis,” where $v_{k,\alpha}$ is expert α ’s loading. The expert decision vector for input x decomposes as:*

$$\sigma(x) = \sum_{k=1}^r c_k(x) \cdot v_k + \eta(x) \quad (42)$$

where $c_k(x) \in \mathbb{R}$ is the “consensus score” along axis k , and $\eta(x)$ is the residual (individual expert noise).

Proposition 25 (Effective Degrees of Freedom). *The effective number of independent expert opinions for input x is:*

$$d_{\text{eff}}(x) = \sum_{k=1}^r \mathbb{1}_{|c_k(x)| > \tau} \quad (43)$$

where τ is a significance threshold. A low d_{eff} indicates strong consensus; a high d_{eff} indicates genuine multi-perspective disagreement.

8.2 Parisi Regime Detection

Regime I Consensus (RS): One dominant eigenvalue: $\lambda_1 \gg \lambda_2 \approx 0$. All experts agree—the audit is trivial.

Regime II Clustered (1RSB): K significant eigenvalues. Experts form K clusters. The audit must report cluster-level conclusions: “within each of the K camps, experts agree; between camps, they disagree.”

Regime III Hierarchical (fRSB): No clear eigenvalue gap. The audit is inherently ambiguous. The report must state: “expert opinions exhibit continuous, fractal disagreement structure; no single consensus exists.”

8.3 Algorithm

Algorithm 1 SCX Hamiltonian Audit Protocol

Require: M experts $\{\mathcal{E}_\alpha\}_{\alpha=1}^M$, task $\mathcal{T}$, threshold δ

Ensure: Audit conclusion with reliability score

- 1: Compute $C_{\alpha\beta} \leftarrow \mathbb{E}_{x \in \mathcal{T}}[\sigma_\alpha(x)\sigma_\beta(x)]$
 - 2: Eigendecomposition: $\{(\lambda_k, v_k)\}_{k=1}^r \leftarrow \text{eig}(C)$
 - 3: Determine Parisi regime:
 - 4: **if** $\frac{\lambda_1}{\sum_k \lambda_k} > 1 - \delta$ **then**
 - 5: **RS:** Consensus; report majority conclusion
 - 6: **else if** $\exists K$ s.t. $\frac{\sum_{k=1}^K \lambda_k}{\sum_k \lambda_k} > 1 - \delta$ and $\frac{\lambda_K}{\lambda_{K+1}} \gg 1$ **then**
 - 7: **1RSB:** K clusters detected; report cluster-level conclusions $\rightarrow$ flag for human review
 - 8: **else**
 - 9: **fRSB:** Hierarchical disagreement; report “inconclusive”; recommend increasing M
 - 10: **end if**
 - 11: Reliability: $R \leftarrow 1 - \exp(-M \cdot I(m_{\text{obs}}))$ (from large deviation theory)
 - 12: **return** conclusion, reliability R , Parisi regime
-

9 Limitations and Open Problems

9.1 Limitations

- L1 Exponential lower bound:** $\Sigma_0(e_\beta) > 0$ implies $\exp(N\Sigma_0)$ grows super-polynomially in N . In practice, the landscape may be simpler—but the bound is a worst-case guarantee. **Implication:** auditing large models may require infeasibly many experts.
- L2 Non-Gaussian landscapes:** $\Sigma_0(e_\beta) > 0$ is the generic case—most loss landscapes are glassy. **Implication:** the exponential barrier is the rule, not the exception.
- L3 Fragility of fRSB:** In the fRSB regime, audit conclusions are sensitive to the exact composition of the expert pool. **Implication:** reproducibility requires documenting the full expert pool—a “which experts” specification.

9.2 Open Problems

In the fRSB regime, the following fundamental question arises:

Proposition 26 (Chaos in Expert Selection). *In the fRSB regime, replacing a single expert $\mathcal{E}$ with a different expert $\mathcal{E}'$ can cause the consensus magnetization to change by an $O(1)$ amount (even as $M \rightarrow \infty$):*

$$\lim_{M \rightarrow \infty} \mathbb{E} m_M(\mathcal{E}) - m_M(\mathcal{E}') > 0 \quad (44)$$

9.3 Connection to Neural Scaling

$$\lim_{\text{width} \rightarrow \infty} \Sigma_0(e) = 0 \quad \text{for all } e > e_{\min} \quad (45)$$

In the infinite-width (NTK) regime, the landscape becomes convex (RS solution), and the exponential barrier vanishes. **Implication:** auditing wide networks is easier—but real networks are not infinitely wide. The effective complexity N_{eff} interpolates between these extremes, governed by $\Sigma_0(e)$.

10 Conclusion

1. We established a rigorous correspondence between the SCX multi-expert auditing framework and the statistical mechanics of disordered systems, with the Hamiltonian $H(\theta) = \mathcal{L}(\theta) + \log Z$ as the central organizing quantity.
2. Each SCX expert corresponds to a Gibbs sample from the Hamiltonian landscape at effective temperature T_α , with coupling between experts mediated by shared training data (the “quenched disorder”).
3. In the SCX consensus model, spontaneous order emerges at a critical temperature $T_c = J$ (where J is expert coupling strength), with a sharp phase transition as $M \rightarrow \infty$.
4. The replica overlap matrix Q_{ab} maps isomorphically to expert-expert correlations $C_{\alpha\beta}$, with identical spectral structure.
5. The Parisi RSB scheme classifies disagreement structures: RS (consensus), 1RSB (clustered disagreement), and fRSB (hierarchical, unauditable disagreement).
6. **Central result:** $M_{\min} \propto \exp(N\Sigma_0(e_\beta))$ —the minimum number of experts grows exponentially with the landscape complexity.

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A Technical Proofs

A.1 Proof of Theorem ?? (Finite-Size Scaling)

$$m = \tanh(\beta J m + \beta h) \quad (46)$$

Set $h = 0$ (no external field). For $m \ll 1$, Taylor expand:

$$m = \beta J m - \frac{(\beta J m)^3}{3} + \mathcal{O}(m^5) \quad (47)$$

$$m(1 - \beta J + \frac{(\beta J)^3 m^2}{3}) = 0 \quad (48)$$

For $1 - \beta J < 0$, i.e., $T < J$ (or $\beta > 1/J$), the nonzero solution is:

$$m = \pm \sqrt{\frac{3(\beta J - 1)}{(\beta J)^3}} \approx \pm \sqrt{31 - \frac{T}{T_c}}, \quad T \rightarrow T_c^- \quad (49)$$

Writing $\beta = \beta_c(1 + \varepsilon)$ with $\varepsilon = (T_c - T)/T \ll 1$. For finite M , the empirical magnetization $m_M = \frac{1}{M} \sum_{\alpha} \sigma_{\alpha}$ has $\text{Var}(m_M) = \frac{1 - m^2}{M}$ (binomial). The signal exceeds noise when m is at least $\sqrt{\text{Var}(m_M)}$:

$$\sqrt{31 - \frac{T}{T_c}} \gtrsim \sqrt{\frac{1}{M}} \implies M \gtrsim \frac{1}{3(1 - T/T_c)} \quad (50)$$

Thus $M_{\min} \propto |T - T_c|^{-1}$, confirming $\nu = 1$. $\square$

A.2 Proof of Theorem ?? (Replica-Expert Isomorphism)

Consider the linear (NTK) regime: $f(x; \theta) = \sum_{i=1}^N \theta_i \phi_i(x)$, where $\phi_i(x)$ are features. For random x :

$$\mathbb{P}(\sigma_{\alpha}(x) = \sigma_{\beta}(x)) = \mathbb{P}(\text{sgn}(\theta_{\alpha} \cdot \phi(x)) = \text{sgn}(\theta_{\beta} \cdot \phi(x))) \quad (51)$$

With $\theta_{\alpha}, \theta_{\beta}$ fixed and $\phi(x)$ isotropic Gaussian, the angle between projections yields:

$$\mathbb{P}(\sigma_{\alpha} = \sigma_{\beta}) = 1 - \frac{1}{\pi} \arccos \frac{\theta_{\alpha} \cdot \theta_{\beta}}{\|\theta_{\alpha}\| \|\theta_{\beta}\|} \quad (52)$$

$$C_{\alpha\beta} = \mathbb{E}[\sigma_{\alpha} \sigma_{\beta}] = \frac{2}{\pi} \arcsin \frac{\theta_{\alpha} \cdot \theta_{\beta}}{\|\theta_{\alpha}\| \|\theta_{\beta}\|} \quad (53)$$

Define $\tilde{Q}_{\alpha\beta} = \frac{\theta_{\alpha} \cdot \theta_{\beta}}{\|\theta_{\alpha}\| \|\theta_{\beta}\|}$. Then $C_{\alpha\beta} = g(\tilde{Q}_{\alpha\beta})$ where $g(z) = \frac{2}{\pi} \arcsin(z)$ is monotone on $[-1, 1]$. As $|\mathcal{X}| \rightarrow \infty$, the empirical C converges to its expectation. The monotonicity of g and the random matrix concentration of $\tilde{Q}$ to $Q_{ab} = \frac{1}{N} \theta_a \cdot \theta_b$ (in the $N \rightarrow \infty$ limit) imply spectral isomorphism. $\square$

A.3 Refined Bound for Theorem ??

Let $\mathcal{N} = \exp(N \Sigma_0(e_{\beta}))$ be the number of significant basins, indexed $k = 1, \dots, \mathcal{N}$. Let p_k be the probability that expert α lands in basin k . Define $I_k^{(M)} = \mathbb{1}_{\{\text{basin } k \text{ is missed by all } M \text{ experts}\}}$. Then $\mathbb{P}(I_k^{(M)} = 1) = (1 - p_k)^M$. The expected number of missed basins is:

$$\mathbb{E} \sum_{k=1}^{\mathcal{N}} I_k^{(M)} = \sum_{k=1}^{\mathcal{N}} (1 - p_k)^M \quad (54)$$

Uniform case ($p_k = 1/\mathcal{N}$ for all k):

$$\mathbb{E}[\text{missed}] = \mathcal{N} \cdot 1 - \frac{1}{\mathcal{N}}^M \approx \mathcal{N} \cdot \exp - \frac{M}{\mathcal{N}} \quad (55)$$

Requiring $\mathbb{E}[\text{missed}] \leq \delta$:

$$\mathcal{N} e^{-M/\mathcal{N}} \leq \delta \implies M \geq \mathcal{N} \log \frac{\mathcal{N}}{\delta} \quad (56)$$

Since $\mathcal{N} \gg 1$ and $\mathcal{N} = \exp(N\Sigma_0)$, this confirms the exponential scaling.

For non-uniform p_k , Bennett’s inequality refines the bound. Let $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(\mathcal{N})}$. Then:

$$\frac{\log \mathcal{N} - \log \delta}{-\log(1 - p_{(1)})} \leq M_{\min} \leq \frac{\log \mathcal{N} - \log \delta}{-\log(1 - p_{(\mathcal{N})})} \quad (57)$$

If $p_{(k)} \propto k^{-\alpha}$ (power-law from Hessian determinant variation), the bounds remain within constant factors. $\square$

B Notation Table

Table 2: Notation

Symbol	Meaning
$H(\theta)$	Hamiltonian (total energy)
$\mathcal{L}(\theta)$	Loss function (potential energy)
$Z(\beta)$	Partition function
$\beta = 1/T$	Inverse temperature
$F(\beta)$	Free energy
M	Number of experts
n	Number of replicas
Q_{ab}	Replica overlap matrix
$C_{\alpha\beta}$	Expert correlation matrix
m_M	Magnetization (consensus)
q_{EA}	Edwards-Anderson order parameter
$q(x)$	Parisi order parameter function
$\Sigma(e)$	Total complexity
$\Sigma_0(e)$	Stable-minima complexity
$\mathcal{N}_N(e)$	Number of critical points at energy e
$\mathcal{H}_{ij}$	Hessian matrix
$M_{\min}$	Minimum number of experts
d_{eff}	Effective degrees of freedom
RS	Replica Symmetric
RSB	Replica Symmetry Breaking
1RSB	One-step RSB
fRSB	Full RSB

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